

H1 - Gradient Descent Hand Trace

Team T16 | M1 - GD | Adil Mutali

Function: $f(x,y) = 2x^2 + 7y^2$; **start:** $x_0 = (4,1)$

1. Derivatives and fixed-step properties

$\text{grad } f = (4x, 14y)$. $H = \text{diag}(4,14)$. Eigenvalues: 4, 14. Condition number $\kappa = 14/4 = 3.5000$. Minimizer $x^* = (0,0)$. Safe range: $0 < \alpha < 2/14 = 0.1429$. Best fixed step: $\alpha^* = 2/(4+14) = 0.1111$. Optimal factor $\rho^* = (14-4)/(14+4) = 0.5556$. $\rho(\alpha) = \max(|1-4\alpha|, |1-14\alpha|)$.

2. Two fixed GD steps, $\alpha = 0.05$

k	$x(k)$	$\text{grad } f(x(k))$	$f(x(k))$
0	(4.0000, 1.0000)	(16.0000, 14.0000)	39.0000
1	(3.2000, 0.3000)	(12.8000, 4.2000)	21.1100
2	(2.5600, 0.0900)	(10.2400, 1.2600)	13.1639

The code reproduces $x(2) = (2.56, 0.09)$ to floating-point precision.

3. One GD-BT step from x_0

$\text{grad } f = (16,14)$, $d = (-16,-14)$, $f(x_0) = 39$, and $\text{grad } f \cdot d = -452$. Armijo RHS = $39 - 0.0452 \alpha$.

Trial α	$f(x_0 + \alpha d)$	Armijo RHS	Decision
1.0000	1471.0000	38.9548	No
0.5000	284.0000	38.9774	No
0.2500	43.7500	38.9887	No
0.1250	11.9375	38.99435	YES

Accepted: $\alpha = 0.1250$ after 3 halvings. $x(1) = (2.0000, -0.7500)$. $f(x(1)) = 11.9375$.

Name: Adil Mutali

Signature: _____